**MATH 111 - Integral Calculus**
**First Semester - 1446 H**
**Solution of the First Exam**
*Dr Tariq A. Alfadhel*

**Question (1):** [9 marks]

1. Use Riemann Sum to evaluate the definite integral $\int_0^2 (x^2+3)\,dx$ . [3]

**Solution :** $[a,b]=[0,2]$ , $f(x)=x^2+3$ .

$$\Delta_x = \frac{b-a}{n} = \frac{2-0}{n} = \frac{2}{n}$$

$$x_k = a + k\ \Delta_x = 0 + k\left(\frac{2}{n}\right) = \frac{2k}{n}$$

$$f\left(x_k\right) = \left(\frac{2k}{n}\right)^2 + 3 = \frac{4k^2}{n^2} + 3$$

$$R_n = \sum_{k=1}^{n} f\left(x_k\right)\ \Delta_x = \sum_{k=1}^{n}\left(\frac{4k^2}{n^2}+3\right)\ \left(\frac{2}{n}\right)$$

$$= \sum_{k=1}^{n}\left(\frac{8k^2}{n^3}+\frac{6}{n}\right) = \sum_{k=1}^{n}\frac{8k^2}{n^3} + \sum_{k=1}^{n}\frac{6}{n}$$

$$= \frac{8}{n^3}\sum_{k=1}^{n}k^2 + \frac{6}{n}\sum_{k=1}^{n}1 = \frac{8}{n^3}\ \frac{n(n+1)(2n+1)}{6} + \frac{6}{n}\ (n)$$

$$= \frac{4}{3}\left(\frac{(n+1)(2n+1)}{n^2}\right) + 6$$

$$\int_0^2 (x^2+3)\,dx = \lim_{n\to\infty} R_n = \lim_{n\to\infty}\left[\frac{4}{3}\left(\frac{(n+1)(2n+1)}{n^2}\right)+6\right]$$

$$= \frac{4}{3}\ (2) + 6 = \frac{8}{3} + 6 = \frac{26}{3}\ .$$

2. Find $F'(x)$, if $F(x) = \int_{\tan\left(\frac{x}{2}\right)}^{3^{5x}} \sqrt{t^2+1}\ dt$ . [2]

**Solution :**

$$F'(x) = \frac{d}{dx}\int_{\tan\left(\frac{x}{2}\right)}^{3^{5x}} \sqrt{t^2+1}\ dt$$

$$= \sqrt{\left(3^{5x}\right)^2+1}\ \left(3^{5x}(5)\ \ln 3\right) - \sqrt{\left(\tan\left(\frac{x}{2}\right)\right)^2+1}\ \left(\sec^2\left(\frac{x}{2}\right)\ \left(\frac{1}{2}\right)\right)$$

$$= 5\ 3^{5x}\ \ln 3\ \sqrt{3^{10x}+1} - \frac{1}{2}\ \sec^2\left(\frac{x}{2}\right)\ \sqrt{\tan^2\left(\frac{x}{2}\right)+1}\ .$$

Find $\dfrac{dy}{dx}$ of the following :

3. $y = \left[\cos^{-1}(3x)\right] \ \log|\csc x - \cot x|$ . [2]

**Solution :**

$$\frac{dy}{dx} = \left(\frac{-1}{\sqrt{1-(3x)^2}}\ (3)\right) \ \log|\csc x - \cot x|$$

$$+ \left[\cos^{-1}(3x)\right] \ \left(\frac{-\csc x \cot x - (-\csc^2)x)}{\csc x - \cot x}\ \frac{1}{\ln 10}\right)$$

$$= \frac{-3\ \log|\csc x - \cot x|}{\sqrt{1-9x^2}} + \frac{\csc x\ \cos^{-1}(3x)}{\ln 10}\ .$$

4. $y = (\cot x)^{\csc x} + 3^{x^2}$ . [2]

**Solution :**

Let $y = f(x) + g(x)$, where $f(x) = (\cot x)^{\csc x}$ and $g(x) = 3^{x^2}$ .

Then $\dfrac{dy}{dx} = y' = f'(x) + g'(x)$

First - $g'(x) = 3^{x^2}\ (2x)\ln 3 = 2x\ 3^{x^2}\ln 3$

Second - Finding $f'(x)$

$f(x) = (\cot x)^{\csc x} \implies \ln|f(x)| = \ln|(\cot x)^{\csc x}| = \csc x\ \ln|\cot x|$

Differentiate both sides.

$$\frac{f'(x)}{f(x)} = (-\csc x\ \cot x)\ \ln|\cot x| + \csc x\left(\frac{-\csc^2 x}{\cot x}\right)$$

$$f'(x) = f(x)\left[-\csc x\ \cot x\ \ln|\cot x| - \frac{\csc^3 x}{\cot x}\right]$$

$$= (\cot x)^{\csc x}\ \left[-\csc x\ \cot x\ \ln|\cot x| - \frac{\csc^3 x}{\cot x}\right]$$

Therefore, $\dfrac{dy}{dx} = (\cot x)^{\csc x}\ \left[-\csc x\ \cot x\ \ln|\cot x| - \dfrac{\csc^3 x}{\cot x}\right] + 2x\ 3^{x^2}\ln 3$

**Question (2):** [16 marks]
Evaluate the following integrals :

1. $\displaystyle\int \left(x^{\frac{2}{5}}\ \sqrt[5]{\sin(x^3)}\right)^5\ dx$ . [2]

**Solution :**

$$\int \left(x^{\frac{2}{5}}\ \sqrt[5]{\sin(x^3)}\right)^5\ dx = \int \left(x^{\frac{2}{5}}\right)^5\ \left(\sqrt[5]{\sin(x^3)}\right)^5\ dx = \int x^2\ \sin(x^3)\ dx$$

$$= \frac{1}{3}\int \sin(x^3)\ (3x^2)\ dx = \frac{1}{3}\ \left(-\cos(x^3)\right) + c = -\frac{1}{3}\ \cos(x^3) + c\ .$$

2. $\displaystyle\int \frac{-1}{\sqrt{e^{6x}-4}}\ dx$ . [2]

**Solution :** .

$$\int \frac{-1}{\sqrt{e^{6x}-4}}\ dx = \int \frac{-1}{\sqrt{(e^{3x})^2-(2)^2}}\ dx = -\int \frac{e^{3x}}{e^{3x}\sqrt{(e^{3x})^2-(2)^2}}\ dx$$

$$= -\frac{1}{3}\int \frac{e^{3x}\ (3)}{e^{3x}\sqrt{(e^{3x})^2-(2)^2}}\ dx = -\frac{1}{3}\left(\frac{1}{2}\ \sec^{-1}\left(\frac{e^{3x}}{2}\right)\right) + c\ .$$

$$= -\frac{1}{6}\ \sec^{-1}\left(\frac{e^{3x}}{2}\right) + c\ .$$

3. $\displaystyle\int_0^1 e^{2\ln x}\ 3^{3x^3}\ dx$ . [2]

**Solution :**

$$\int_0^1 e^{2\ln x}\ 3^{3x^3}\ dx = \int_0^1 e^{\ln x^2}\ 3^{3x^3}\ dx = \int_0^1 x^2\ 3^{3x^3}\ dx$$

$$= \frac{1}{9}\int_0^1 3^{3x^3}(9x^2)\ dx = \frac{1}{9}\left[\frac{3^{3x^3}}{\ln 3}\right]_0^1 = \frac{1}{9}\left[\frac{3^{3(1)^3}}{\ln 3} - \frac{3^{3(0)^3}}{\ln 3}\right]$$

$$= \frac{1}{9}\left[\frac{3^3}{\ln 3} - \frac{3^0}{\ln 3}\right] = \frac{1}{9}\ \left(\frac{27-1}{\ln 3}\right) = \frac{26}{9\ln 3}\ .$$

4. $\displaystyle\int \frac{x+1}{\sqrt{1-x^2}}\ dx$ . [2]

**Solution :**

$$\int \frac{x+1}{\sqrt{1-x^2}}\ dx = \int \left[\frac{x}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}}\right]\ dx$$

$$= \int \frac{x}{\sqrt{1-x^2}}\ dx + \int \frac{1}{\sqrt{1-x^2}}\ dx$$

$$= -\frac{1}{2}\int \left(1-x^2\right)^{-\frac{1}{2}}\ (-2x)\ dx + \int \frac{1}{\sqrt{1-x^2}}\ dx$$

$$= -\frac{1}{2}\ \frac{\left(1-x^2\right)^{\frac{1}{2}}}{\frac{1}{2}} + \sin^{-1}x + c = -\sqrt{1-x^2} + \sin^{-1}x + c\ .$$

5. $\displaystyle\int \frac{\sec\left(\ln(x^2)\right)}{x}\,dx$ . [2]

**Solution :**

$$\int \frac{\sec\left(\ln(x^2)\right)}{x}\,dx = \int \sec\left(2\ln|x|\right)\ \frac{1}{x}\,dx$$

$$= \frac{1}{2}\int \sec\left(2\ln|x|\right)\ \frac{2}{x}\,dx = \frac{1}{2}\ln\left|\sec\left(2\ln|x|\right)+\tan\left(2\ln|x|\right)\right| + c$$

6. $\displaystyle\int \frac{\tan\left(\sqrt[3]{x}\right)}{x^{\frac{2}{3}}}\,dx$ . [2]

**Solution :**

$$\int \frac{\tan\left(\sqrt[3]{x}\right)}{x^{\frac{2}{3}}}\,dx = \int \tan\left(x^{\frac{1}{3}}\right)\ x^{-\frac{2}{3}}\,dx$$

$$= 3\int \tan\left(x^{\frac{1}{3}}\right)\left(\frac{1}{3}\,x^{-\frac{2}{3}}\right)\,dx = 3\ln\left|\sec\left(\sqrt[3]{x}\right)\right| + c$$

7. $\displaystyle\int \frac{\left(1+\sec^{-1}x\right)^5}{\sqrt{x^4-x^2}}\,dx$ . [2]

**Solution :**

$$\int \frac{\left(1+\sec^{-1}x\right)^5}{\sqrt{x^4-x^2}}\,dx = \int \frac{\left(1+\sec^{-1}x\right)^5}{\sqrt{x^2\left(x^2-1\right)}}\,dx$$

$$= \int \frac{\left(1+\sec^{-1}x\right)^5}{|x|\sqrt{x^2-1}}\,dx = \int \left(1+\sec^{-1}x\right)^5\left(\frac{1}{x\sqrt{x^2-1}}\right)\,dx$$

$$= \frac{\left(1+\sec^{-1}x\right)^6}{6} + c \text{ , where } x > 0 \text{ .}$$

8. $\displaystyle\int \cos x\ \left(\sin^2 x\right)^{-1}\,dx$ . [2]

**First solution :**

$$\int \cos x\ \left(\sin^2 x\right)^{-1}\,dx = \int (\sin x)^{-2}\cos x\,dx = \frac{(\sin x)^{-1}}{-1} + c = -\csc x + c$$

**Second solution :**

$$\int \cos x\ \left(\sin^2 x\right)^{-1}\,dx = \int \frac{\cos x}{\sin^2 x}\,dx = \int \frac{1}{\sin x}\frac{\cos x}{\sin x}\,dx$$

$$= \int \csc x\ \cot x\,dx = -\csc x + c\ .$$

**MATH 111 - Integral Calculus**
**First Semester - 1446 H**
**Solution of the Second Exam**
*Dr Tariq A. Alfadhel*

**Question (1):** [4 marks]
Find $\frac{dy}{dx}$ of the following :

1. $y = csch\left(\frac{2}{x}\right) + \tanh^{-1}\left(e^{5x}\right)$ . [2]

   **Solution :**

   $$\frac{dy}{dx} = -csch\left(\frac{2}{x}\right)\coth\left(\frac{2}{x}\right)\left(\frac{-2}{x^2}\right) + \frac{1}{1-\left(e^{5x}\right)^2}\left(e^{5x}\ (5)\right)$$

   $$= \frac{2}{x^2}\ csch\left(\frac{2}{x}\right)\coth\left(\frac{2}{x}\right) + \frac{5e^{5x}}{1-e^{10x}}\ .$$

2. $y = \sinh^{-1}\left(\sqrt{x}\ \right) + \coth(4x^2)$ . [2]

   **Solution :**

   $$\frac{dy}{dx} = \frac{1}{\sqrt{1+\left(\sqrt{x}\right)^2}}\left(\frac{1}{2\sqrt{x}}\right) + \left(-csch^2(4x^2)\ (8x)\right)$$

   $$= \frac{1}{2\sqrt{x}\sqrt{1+x}} - 8x\ csch^2(4x^2)\ .$$

**Question (2):** [21 marks]
Evaluate the following integrals :

1. $\int \frac{sech^2(x^{-2})}{x^3}\ dx$ . [2]

   **Solution :**

   $$\int \frac{sech^2(x^{-2})}{x^3}\ dx = \int sech^2(x^{-2})\ x^{-3}\ dx$$

   $$= \frac{1}{-2}\int sech^2(x^{-2})\ \left(-2x^{-3}\right)\ dx = -\frac{1}{2}\tanh(x^{-2}) + c$$

2. $\int \frac{3}{\sqrt{x^2+6x-16}}\ dx$ . [3]

   **Solution :** By completing the square.

   $$x^2+6x-16 = (x^2+6x+9) - 16 - 9 = (x+3)^2 - (5)^2\ .$$

$$\int \frac{3}{\sqrt{x^2+6x-16}}\,dx = 3\int \frac{1}{\sqrt{(x+3)^2-(5)^2}}\,dx$$

$$= 3\cosh^{-1}\left(\frac{x+3}{5}\right) + c\ .$$

3. $\int e^x\ \cosh 2x\ dx$ . [2]

**Solution :**

$$\int e^x\ \cosh 2x\ dx = \int e^x\left(\frac{e^{2x}+e^{-2x}}{2}\right)\ dx = \int\left(\frac{e^{3x}+e^{-x}}{2}\right)\ dx$$

$$\int\left(\frac{e^{3x}}{2}+\frac{e^{-x}}{2}\right)\ dx = \frac{1}{2}\ \frac{1}{3}\int e^{3x}(3)\ dx + \frac{1}{2}\ \frac{1}{-1}\int e^{-x}(-1)\ dx$$

$$= \frac{1}{6}\ e^{3x} - \frac{1}{2}\ e^{-x} + c = \frac{e^{3x}}{6} - \frac{e^{-x}}{2} + c\ .$$

4. $\int (3x-2)\ \sinh x\ dx$ . [2]

**Solution :** Using integration by parts.

| | |
|---|---|
| $u = 3x-2$ | $dv = \sinh x\ dx$ |
| $du = 3\ dx$ | $v = \cosh x$ |

$$\int (3x-2)\ \sinh x\ dx = (3x-2)\ \cosh x - \int 3\cosh x\ dx$$

$$= (3x-2)\ \cosh x - 3\int \cosh x\ dx = (3x-2)\ \cosh x - 3\sinh x + c\ .$$

5. $\int x^{-2}\ \ln x\ dx$ . [2]

**Solution :** Using integration by parts.

| | |
|---|---|
| $u = \ln x$ | $dv = x^{-2}\ dx$ |
| $du = \frac{1}{x}\ dx$ | $v = \frac{x^{-1}}{-1} = \frac{-1}{x}$ |

$$\int x^{-2}\ \ln x\ dx = \frac{-1}{x}\ln x - \int \frac{-1}{x}\ \frac{1}{x}\ dx$$

$$= \frac{-\ln x}{x} - \int \frac{-1}{x^2}\ dx = \frac{-\ln x}{x} - \frac{1}{x} + c\ .$$

6. $\int \sin^4 x \ \cos^5 x \ dx$ . [2]

**Solution :**

Using the substitution $u = \sin x$ .

Hence $du = \cos x \ dx$ .

$$\int \sin^4 x \ \cos^5 x \ dx = \int \sin^4 x \ \cos^4 x \ \cos x \ dx = \int \sin^4 x \ \left(\cos^2 x\right)^2 \cos x \ dx$$

$$= \int \sin^4 x \ \left(1 - \sin^2 x\right)^2 \cos x \ dx = \int u^4 \left(1 - u^2\right)^2 \ du = \int u^4 \left(1 - 2u^2 + u^4\right) \ du$$

$$= \int \left(u^4 - 2u^6 + u^8\right) \ du = \frac{u^5}{5} - 2 \ \frac{u^7}{7} + \frac{u^9}{9} + c$$

$$= \frac{\sin^5 x}{5} - 2 \ \frac{\sin^7 x}{7} + \frac{\sin^9 x}{9} + c$$

7. $\int \dfrac{\sqrt{x^2 - 9}}{x} \ dx$ . [3]

**Solution :** Using trigonometric substitutions.

Put $x = 3 \sec\theta \implies \sec\theta = \dfrac{x}{3} \implies \cos\theta = \dfrac{3}{x}$ .

$dx = 3 \sec\theta \ \tan\theta \ d\theta$ .

$$\sqrt{x^2 - 9} = \sqrt{9 \sec^2\theta - 9} = \sqrt{9\left(\sec^2\theta - 1\right)} = \sqrt{9 \tan^2\theta} = 3 \tan\theta$$

$$= \int \frac{\sqrt{x^2 - 9}}{x} \ dx = \int \frac{(3 \tan\theta) \ (3 \sec\theta \ \tan\theta)}{3 \sec\theta} \ d\theta = 3 \int \tan^2\theta \ d\theta$$

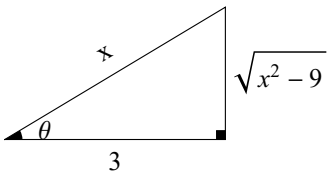
$$= 3 \int \left(\sec^2\theta - 1\right) \ d\theta = 3 \left(\tan\theta - \theta\right) + c = 3 \tan\theta - 3\theta + c$$

$\cos\theta = \dfrac{3}{x}$ .

From the triangle :

$\tan\theta = \dfrac{\sqrt{x^2 - 9}}{3}$

$\theta = \sec^{-1}\left(\dfrac{x}{3}\right)$




$$\int \frac{\sqrt{x^2 - 9}}{x} \ dx = 3 \ \left(\frac{\sqrt{x^2 - 9}}{3}\right) - 3 \sec^{-1}\left(\frac{x}{3}\right) + c$$

$$= \sqrt{x^2 - 9} - 3 \sec^{-1}\left(\frac{x}{3}\right) + c \ .$$

8. $\int \dfrac{5x^2+x+8}{x^3+4x}\,dx$ . [3]

**Solution :** Using the method of partial fractions.

$$\frac{5x^2+x+8}{x^3+4x} = \frac{5x^2+x+8}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

$$\frac{5x^2+x+8}{x(x^2+4)} = \frac{A(x^2+4)}{x(x^2+4)} + \frac{x\,(Bx+C)}{x(x^2+4)}$$

$$5x^2+x+8 = A(x^2+4) + x(Bx+C)$$

$$5x^2+x+8 = Ax^2+4A+Bx^2+Cx = (A+B)x^2+Cx+4A$$

By comparing the coefficients of the two polynomials in each side :

$$\begin{array}{lll} A+B=5 & \longrightarrow & (1) \\ C=1 & \longrightarrow & (2) \\ 4A=8 & \longrightarrow & (3) \end{array}$$

From equation (3) : $4A=8 \implies A=\dfrac{8}{4}=2$ .

From equation (1) : $2+B=0 \implies B=3$ .

$$\int \frac{5x^2+x+8}{x^3+4x}\,dx = \int \left(\frac{2}{x}+\frac{3x+1}{x^2+4}\right)\,dx$$

$$= \int \frac{2}{x}\,dx + \int \frac{3x}{x^2+4}\,dx + \int \frac{1}{x^2+4}\,dx$$

$$= 2\int \frac{1}{x}\,dx + \frac{3}{2}\int \frac{2x}{x^2+4}\,dx + \int \frac{1}{x^2+(2)^2}\,dx$$

$$= 2\ln|x| + \frac{3}{2}\ln(x^2+4) + \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + c\ .$$

9. $\int \dfrac{dx}{\sqrt[4]{x}+\sqrt{x}}$ . [2]

**Solution :** $\displaystyle\int \frac{dx}{\sqrt[4]{x}+\sqrt{x}} = \int \frac{1}{x^{\frac{1}{4}}+x^{\frac{1}{2}}}\,dx$

Using the substitution $x=u^4$, then $u=x^{\frac{1}{4}}$ .

$dx = 4u^3\,du$ .

$$\int \frac{dx}{x^{\frac{1}{4}}+x^{\frac{1}{2}}} = \int \frac{4u^3}{(u^4)^{\frac{1}{4}}+(u^4)^{\frac{1}{2}}}\,du = \int \frac{4u^3}{u+u^2}\,du$$

$$= \int \frac{4u^3}{u(1+u)}\,du = 4\int \frac{u^2}{u+1}\,du$$

Using long division of polynomials :

$$4\int \frac{u^2}{u+1}\,du = 4\int \left(u-1+\frac{1}{u+1}\right)\,du$$

$$= 4\left(\frac{u^2}{2}-u+\ln|u+1|\right)+c = 2u^2-4u+4\ln|u-1|+c$$

$$\int \frac{dx}{x^{\frac{1}{4}}+x^{\frac{1}{2}}} = 2\left(x^{\frac{1}{4}}\right)^2-4x^{\frac{1}{4}}+4\ln\left|x^{\frac{1}{4}}+1\right|+c$$

$$= 2x^{\frac{1}{2}}-4x^{\frac{1}{4}}+4\ln\left|x^{\frac{1}{4}}+1\right|+c\ .$$

**MATH 111 - Integral Calculus**
**First Semester - 1446 H**
**Solution of the Final Exam**
*Dr Tariq A. Alfadhel*

**Question (1):** [7 marks]

1. Find the value of $c$ that satisfies the mean value theorem of the definite integral for the function $f(x) = 4x - x^2$ on the interval $[0, 3]$. [3]

   **Solution :** Using the formula $(b-a)\ f(c) = \int_a^b f(x)\ dx$ .

   $$(3-0)\ (4c - c^2) = \int_0^3 (4x - x^2)\ dx = \left[2x^2 - \frac{x^3}{3}\right]_0^3$$

   $$3(4c - c^2) = \left(2(3)^2 - \frac{(3)^3}{3}\right) - \left(2(0)^2 - \frac{(0)^3}{3}\right) = 18 - 9 = 9$$

   $$3(4c - c^2) = 9 \implies 4c - c^2 = 3 \implies c^2 - 4c + 3 = 0$$

   $$\implies (c-3)(c-1) = 0 \implies c = 3\ ,\ c = 1.$$

   Note that $c = 1 \in (0, 3)$ while $c = 3 \notin (0, 3)$.

   The desired value is $c = 1$ .

2. Find $F'(x)$, if $F(x) = \int_{\sec x}^{3^{2x}} \sqrt{2t^2 - 1}\ dt$ . [2]

   **Solution :**

   $$F'(x) = \frac{d}{dx} \int_{\sec x}^{3^{2x}} \sqrt{2t^2 - 1}\ dt$$

   $$= \sqrt{2\left(3^{2x}\right)^2 - 1}\ \left(3^{2x}(2)\ \ln 3\right) - \sqrt{2\left(\sec x\right)^2 - 1}\ \left(\sec x\ \tan x\right)$$

   $$= 2\ln 3\left(3^{2x}\right)\sqrt{2\ \left(3^{4x}\right) - 1} - \sec x \tan x \sqrt{2\sec^2 x - 1}\ .$$

3. Find $y'$ if $y = \cosh^{-1}\left(3^{x^2-1}\right) + \log|\tanh(2x)|$ . [2]

   **Solution :**

   $$y' = \frac{1}{\sqrt{\left(3^{x^2-1}\right)^2 - 1}}\ \left(3^{x^2-1}\ (2x)\ \ln 3\right) + \frac{sech^2(2x)\ (2)}{\tanh(2x)}\ \frac{1}{\ln 10}$$

   $$= \frac{2x \ln 3\ \left(3^{x^2-1}\right)}{\sqrt{3^{2x^2-2} - 1}} + \frac{2\ sech^2(2x)}{\tanh(2x)\ \ln 10}\ .$$

**Question (2):** [14 marks]
Evaluate the following integrals :

1. $\int \dfrac{2}{\sqrt{-x^2+6x-8}}\,dx$ . [3]

**Solution :** By completing the square.

$-x^2+6x-8 = -(x^2-6x)-8 = -(x^2-6x+9)-8+9$

$= -(x-3)^2+1 = 1-(x-3)^2$ .

$$\int \frac{2}{\sqrt{-x^2+6x-8}}\,dx = 2\int \frac{1}{\sqrt{1-(x-3)^2}}\,dx = 2\sin^{-1}(x-3)+c\ .$$

2. $\int x^{-3}\ \ln|x|\ dx$ . [3]

**Solution :** Using integration by parts .

$u = \ln x \qquad dv = x^{-3}\,dx$

$du = \dfrac{1}{x}\,dx \qquad v = \dfrac{x^{-2}}{-2} = \dfrac{1}{-2x^2}$

$$\int x^{-3}\ \ln x\ dx = \frac{1}{-2x^2}\ln x - \int \frac{1}{-2x^2}\ \frac{1}{x}\ dx$$

$$= \frac{\ln x}{-2x^2} + \frac{1}{2}\int x^{-3}\,dx = \frac{\ln x}{-2x^2} + \frac{1}{2}\left(\frac{x^{-2}}{-2}\right)+c = \frac{\ln x}{-2x^2} - \frac{1}{4x^2}+c\ .$$

3. $\int \dfrac{\sqrt{x^2+9}}{x^4}\,dx$ . [3]

**Solution :** Using trigonometric substitutions.

Put $x = 3\tan\theta \implies \tan\theta = \dfrac{x}{3}$ .

$dx = 3\sec^2\theta\ d\theta$ .

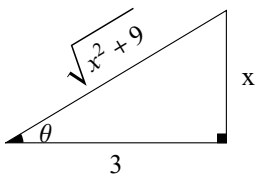
$$\sqrt{x^2+9} = \sqrt{9\tan^2\theta+9} = \sqrt{9\left(\tan^2\theta+1\right)} = \sqrt{9\sec^2\theta} = 3\sec\theta\ .$$

$$\int \frac{\sqrt{x^2+9}}{x^4}\,dx = \int \frac{3\sec\theta\ 3\sec^2\theta}{(3\tan\theta)^4}\,d\theta = \frac{3^2}{3^4}\int \frac{\sec^3\theta}{\tan^4\theta}\,d\theta$$

$$= \frac{1}{9}\int \frac{\cos^4\theta}{\sin^4\theta\ \cos^3\theta}\,d\theta = \frac{1}{9}\int (\sin\theta)^{-4}\ \cos\theta\ d\theta = \frac{1}{9}\ \frac{(\sin\theta)^{-3}}{-3}+c$$

$\tan\theta = \dfrac{x}{3}$ .

From the triangle :
$\sin\theta = \dfrac{x}{\sqrt{x^2+9}}$



$$\int \frac{\sqrt{x^2+9}}{x^4}\,dx = \frac{1}{-27}\left(\frac{x}{\sqrt{x^2+9}}\right)^{-3} + c\ .$$

4. $\displaystyle\int \frac{3x+1}{x^3+x}\,dx$ . [3]

**Solution :** Using the method of partial fractions.

$$\frac{3x+1}{x^3+x} = \frac{3x+1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$$

$$\frac{3x+1}{x(x^2+1)} = \frac{A(x^2+1)}{x(x^2+1)} + \frac{x\,(Bx+C)}{x(x^2+1)}$$

$$3x+1 = A(x^2+1) + x(Bx+C)$$

$$3x+1 = Ax^2 + A + Bx^2 + Cx = (A+B)x^2 + Cx + A$$

By comparing the coefficients of the two polynomials in each side :

$$\begin{array}{lll} A+B=0 & \longrightarrow & (1) \\ C=3 & \longrightarrow & (2) \\ A=1 & \longrightarrow & (3) \end{array}$$

From equation (1) : $1+B=0 \implies B=-1$ .

$$\int \frac{3x+1}{x^3+x}\,dx = \int \left(\frac{1}{x} + \frac{-x+3}{x^2+1}\right)\,dx$$

$$= \int \frac{1}{x}\,dx + \int \frac{-x}{x^2+1}\,dx + \int \frac{3}{x^2+1}\,dx$$

$$= \int \frac{1}{x}\,dx - \frac{1}{2}\int \frac{2x}{x^2+1}\,dx + 3\int \frac{1}{x^2+(1)^2}\,dx$$

$$= \ln|x| - \frac{1}{2}\ln(x^2+1) + 3\tan^{-1}(x) + c\ .$$

5. $\displaystyle\int \frac{1}{\sqrt{x}\ (1+x)}\,dx$ . [2]

**Solution :** Using the substitution $u=\sqrt{x} \implies u^2 = x$ .

$2u\ du = dx$ .

$$\int \frac{1}{\sqrt{x}\ (1+x)}\,dx = \int \frac{2u}{u(1+u^2)}\,du = 2\int \frac{1}{1+u^2}\,du$$

$$= 2\tan^{-1}(u) + c = 2\tan^{-1}\left(\sqrt{x}\right) + c\ .$$

**Question (3):** [19 marks]

1. Evaluate the limit $\lim\limits_{x\to\infty} e^{-x^3}\ (x^4+1)$ . [2]

   **Solution :**

   $\lim\limits_{x\to\infty} e^{-x^3}\ (x^4+1) \qquad (0.\infty)$

   $\lim\limits_{x\to\infty} e^{-x^3}\ (x^4+1) = \lim\limits_{x\to\infty} \dfrac{x^4+1}{e^{x^3}} \qquad \left(\dfrac{\infty}{\infty}\right)$

   Using L'Hôpital's rule.

   $\lim\limits_{x\to\infty} \dfrac{4\ x^3}{e^{x^3}\ (3x^2)} = \lim\limits_{x\to\infty} \dfrac{4x}{3e^{x^3}} \qquad \left(\dfrac{\infty}{\infty}\right)$

   Using L'Hôpital's rule.

   $\lim\limits_{x\to\infty} \dfrac{4}{3e^{x^3}(3x^2)} = \lim\limits_{x\to\infty} \dfrac{4}{9x^2\ e^{x^3}} = 0$ .

   Therefore, $\lim\limits_{x\to\infty} e^{-x^3}\ (x^4+1) = 0$ .

   Note that $\lim\limits_{x\to\infty} e^{x^3} = \infty$ and $\lim\limits_{x\to\infty} 9x^2 = \infty$ .

2. Discus whether the improper integral $\displaystyle\int_2^{\infty} \frac{1}{x\,(\ln x)^2}\ dx$ converges or diverges. [3]

   **Solution :**

   $$\int_2^{\infty} \frac{1}{x\,(\ln x)^2}\ dx = \lim_{t\to\infty} \int_2^{t} \frac{1}{x\,(\ln x)^2}\ dx = \lim_{t\to\infty} \int_2^{t} (\ln x)^{-2}\,\frac{1}{x}\ dx$$

   $$= \lim_{t\to\infty} \left[\frac{(\ln x)^{-1}}{-1}\right]_2^{t} = \lim_{t\to\infty} \left[\frac{-1}{\ln x}\right]_2^{t} = \lim_{t\to\infty} \left[\frac{-1}{\ln t} - \frac{-1}{\ln 2}\right]$$

   $$= 0 - \frac{-1}{\ln 2} = \frac{1}{\ln 2}\ .$$

   Hence, the improper integral converges.

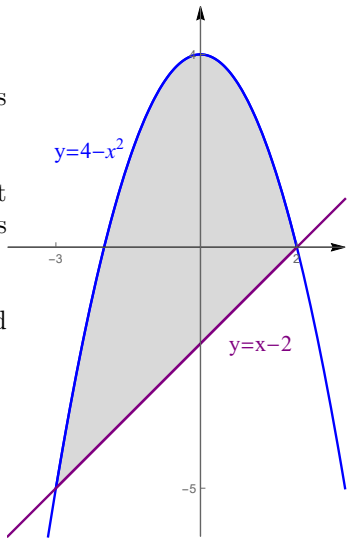
3. Sketch the region region bounded by the curves $y = 4 - x^2$ and $y = x - 2$, and find its area. [3]

   **Solution :**

$y = 4 - x^2$ represents a parabola opens downwards, and its vertex is $(0, 4)$ .

$y = x - 2$ represents a straight line passing through $(0, -2)$ , and its slope equals 1.

Points of intersection of $y = 4 - x^2$ and $y = x - 2$:
$x - 2 = 4 - x^2 \implies x^2 + x - 6 = 0$
$\implies (x + 3)(x - 2) = 0$
$\implies x = -3 \ , \ x = 2.$




$$\mathbf{A} = \int_{-3}^{2} \left[(4 - x^2) - (x - 2)\right] \, dx = \int_{-3}^{2} (4 - x^2 - x + 2) \, dx$$

$$= \int_{-3}^{2} (-x^2 - x + 6) \, dx = \left[-\frac{x^3}{3} - \frac{x^2}{2} + 6x\right]_{-3}^{2}$$

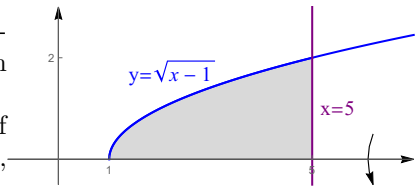
$$= \left(-\frac{(2)^3}{3} - \frac{(2)^2}{2} + 6(2)\right) - \left(-\frac{(-3)^3}{3} - \frac{(-3)^2}{2} + 6(-3)\right)$$

$$= \left(\frac{-8}{3} - 2 + 12\right) - \left(9 - \frac{9}{2} - 18\right) = 19 - \frac{8}{3} + \frac{9}{2} = \frac{125}{6} \ .$$

4. Sketch the region bounded bu the curves $y = \sqrt{x - 1}$ , $y = 0$ and $x = 5$, and find the volume of the solid generated by revolving this region about the $x-$axis. [3]

**Solution :**

$y = 0$ represents the $x$-axis.
$x = 5$ represents a straight line parallel to the $y$-axis, and passing through $(5, 0)$.
$y = \sqrt{x - 1}$ represents the upper half of the parabola $x = y^2$ opens to the right, and its vertex is $(0, 1)$.




Using the disk method.

$$\mathbf{V} = \pi \int_{1}^{5} \left(\sqrt{x - 1}\right)^2 \, dx = \pi \int_{1}^{5} (x - 1) \, dx = \left[\frac{x^2}{2} - x\right]_{1}^{5}$$

$$= \left(\frac{25}{2} - 5\right) - \left(\frac{1}{2} - 1\right) = \frac{25}{2} - 5 - \frac{1}{2} + 1 = 12 - 4 = 8 \ .$$

5. Find the arc length of the function $y = \ln|\sec x|$ from $x = 0$ to $x = \dfrac{\pi}{4}$. [3]

**Solution :**

$$y' = \frac{\sec x \tan x}{\sec x} = \tan x \ .$$

$$\mathbf{L} = \int_0^{\frac{\pi}{4}} \sqrt{1 + (\tan x)^2}\ dx = \int_0^{\frac{\pi}{4}} \sqrt{1 + \tan^2 x}\ dx = \int_0^{\frac{\pi}{4}} \sqrt{\sec^2 x}\ dx$$

$$= \int_0^{\frac{\pi}{4}} |\sec x|\ dx = \int_0^{\frac{\pi}{4}} \sec x\ dx = [\ln|\sec x + \tan x|]_0^{\frac{\pi}{4}}$$

$$= \ln\left|\sec\left(\frac{\pi}{4}\right) + \tan\left(\frac{\pi}{4}\right)\right| - \ln|\sec(0) + \tan(0)|$$

$$= \ln\left|\sqrt{2} + 1\right| - \ln|1 + 0| = \ln\left|1 + \sqrt{2}\right| \ .$$

6. Convert the polar equation $r = \dfrac{1}{\sin\theta - 2\cos\theta}$ into a Cartesian equation. [1]

**Solution :**

$$r = \frac{1}{\sin\theta - 2\cos\theta} \implies r(\sin\theta - 2\cos\theta) = 1$$

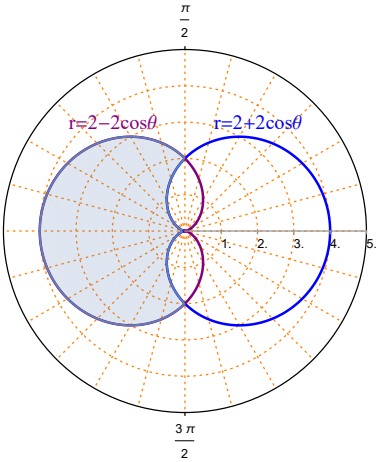
$$\implies r\sin\theta - 2(r\cos\theta) = 1 \implies y - 2x = 1 \implies y = 2x + 1 \ .$$

7. Sketch the region inside the graph of the polar equation $r = 2 - 2\cos\theta$ and outside the graph of $r = 2 + 2\cos\theta$. Then compute its area. [4]

**Solution :**

Points of intersection of $r = 2 - 2\cos\theta$ and $r = 2 + 2\cos\theta$ :
$2 + 2\cos\theta = 2 - 2\cos\theta$
$\implies 4\cos\theta = 0$
$\implies \cos\theta = 0$
$\implies \theta = \dfrac{\pi}{2}\ ,\ \theta = \dfrac{3\pi}{2}$ .

Note that the shaded region is symmetric with respect to the polar axis.




$$\mathbf{A} = 2\left(\frac{1}{2}\int_{\frac{\pi}{2}}^{\pi}\left[(2 - 2\cos\theta)^2 - (2 + 2\cos\theta)^2\right]\ d\theta\right)$$

$$= \int_{\frac{\pi}{2}}^{\pi} \left[4 - 8\cos\theta + 4\cos^2\theta - \left(4 + 8\cos\theta + 4\cos^2\theta\right)\right] \, d\theta$$

$$= \int_{\frac{\pi}{2}}^{\pi} \left(4 - 8\cos\theta + 4\cos^2\theta - 4 - 8\cos\theta - 4\cos^2\theta\right) \, d\theta$$

$$= \int_{\frac{\pi}{2}}^{\pi} -16\cos\theta \, d\theta = -16\left[\sin\theta\right]_{\frac{\pi}{2}}^{\pi}$$

$$= -16\left[\sin(\pi) - \sin\left(\frac{\pi}{2}\right)\right] = -16(0-1) = 16 \ .$$

**MATH 111 - Integral Calculus**
**Second Semester - 1446 H**
**Solution of the First Exam**
*Dr Tariq A. Alfadhel*

**Question (1):** [9 marks]

1. Use Riemann Sum to evaluate the definite integral $\int_0^4 (2x^2+1)\ dx$ . [3]

**Solution :** $[a,b]=[0,4]$ , $f(x)=2x^2+1$ .

$$\Delta_x = \frac{b-a}{n} = \frac{4-0}{n} = \frac{4}{n}$$

$$x_k = a + k\ \Delta_x = 0 + k\left(\frac{4}{n}\right) = \frac{4k}{n}$$

$$f(x_k) = 2\left(\frac{4k}{n}\right)^2 + 1 = 2\left(\frac{16k^2}{n^2}\right) + 1 = \frac{32k^2}{n^2} + 1$$

$$R_n = \sum_{k=1}^{n} f(x_k)\ \ \Delta_x = \sum_{k=1}^{n}\left(\frac{32k^2}{n^2}+1\right)\ \ \left(\frac{4}{n}\right)$$

$$= \sum_{k=1}^{n}\left(\frac{128k^2}{n^3} + \frac{4}{n}\right) = \sum_{k=1}^{n}\frac{128k^2}{n^3} + \sum_{k=1}^{n}\frac{4}{n}$$

$$= \frac{128}{n^3}\sum_{k=1}^{n}k^2 + \frac{4}{n}\sum_{k=1}^{n}1 = \frac{128}{n^3}\ \frac{n(n+1)(2n+1)}{6} + \frac{4}{n}\ (n)$$

$$= \frac{64}{3}\left(\frac{(n+1)(2n+1)}{n^2}\right) + 4$$

$$\int_0^4 (2x^2+1)\ dx = \lim_{n\to\infty} R_n = \lim_{n\to\infty}\left[\frac{64}{3}\left(\frac{(n+1)(2n+1)}{n^2}\right) + 4\right]$$

$$= \frac{64}{3}\ (2) + 4 = \frac{128}{3} + 4 = \frac{140}{3}\ .$$

2. Find $F'(x)$, if $F(x) = \int_{\tan\left(\sqrt[3]{x}\right)}^{3^{5x}} \sqrt{t^2+1}\ dt$ . [2]

**Solution :**

$$F'(x) = \frac{d}{dx}\int_{\tan\left(\sqrt[3]{x}\right)}^{3^{5x}} \sqrt{t^2+1}\ dt$$

$$= \sqrt{\left(3^{5x}\right)^2+1}\ \left(3^{5x}(5)\ \ln 3\right) - \sqrt{\left(\tan\left(\sqrt[3]{x}\right)\right)^2+1}\ \left(\sec^2\left(\sqrt[3]{x}\right)\ \ \left(\frac{1}{3}\ x^{-\frac{2}{3}}\right)\right)$$

$$= 5\ 3^{5x}\ \ln 3\ \sqrt{3^{10x}+1} - \frac{\sec^2\left(\sqrt[3]{x}\right)\ \sqrt{\tan^2\left(\sqrt[3]{x}\right)+1}}{3\ x^{\frac{2}{3}}}\ .$$

Find $\dfrac{dy}{dx}$ of the following :

3. $y = \left[\csc^{-1}\left(\dfrac{3}{x}\right)\right] \ \log|\sec(2x)+\tan(2x)|$ . [2]

**Solution :**

$$\frac{dy}{dx} = \left(\frac{-1}{\left(\frac{3}{x}\right)\sqrt{\left(\frac{3}{x}\right)^2-1}}\ \left(\frac{-3}{x^2}\right)\right)\ \log|\sec(2x)+\tan(2x)|$$

$$+\left[\csc^{-1}\left(\frac{3}{x}\right)\right]\ \left(\frac{\sec(2x)\tan(2x)\ (2)+\sec^2(2x)\ (2)}{\sec(2x)+\tan(2x)}\ \frac{1}{\ln 10}\right)$$

$$= \frac{3\ \log|\sec(2x)+\tan(2x)|}{3x\sqrt{\frac{9}{x^2}-1}} + \frac{2\sec(2x)\csc^{-1}\left(\frac{3}{x}\right)}{\ln 10}\ .$$

4. $y = (\cos x)^{\csc x} + 3^{2x^2}$ . [2]

**Solution :**

Let $y = f(x)+g(x)$, where $f(x) = (\cos x)^{\csc x}$ and $g(x) = 3^{2x^2}$ .

Then $\dfrac{dy}{dx} = y' = f'(x)+g'(x)$

First : $g'(x) = 3^{2x^2}\ (4x)\ln 3 = 4x\ 3^{x^2}\ln 3$

Second : Finding $f'(x)$

$f(x) = (\cos x)^{\csc x} \implies \ln|f(x)| = \ln|(\cos x)^{\csc x}| = \csc x\ \ln|\cos x|$

Differentiate both sides.

$$\frac{f'(x)}{f(x)} = (-\csc x\ \cot x)\ \ln|\cos x| + \csc x\left(\frac{-\sin x}{\cos x}\right)$$

$$f'(x) = f(x)\left[-\csc x\ \cot x\ \ln|\cos x| - \frac{\csc x\ \sin x}{\cos x}\right]$$

$$= (\cos x)^{\csc x}\ \left[-\csc x\ \cos x\ \ln|\cos x| - \frac{1}{\cos x}\right]$$

Therefore, $\dfrac{dy}{dx} = (\cos x)^{\csc x}\ [-\csc x\ \cos x\ \ln|\cos x| - \sec x] + 4x\ 3^{x^2}\ln 3$

**Question (2):** [16 marks]
Evaluate the following integrals :

1. $\displaystyle\int \left(x^{\frac{2}{3}}\ \sqrt[3]{\sec(x^3)}\right)^3\ dx$ . [2]

**Solution :**

$$\int \left(x^{\frac{2}{3}} \sqrt[3]{\sec(x^3)}\right)^3 \, dx = \int \left(x^{\frac{2}{3}}\right)^3 \left(\sqrt[3]{\sec(x^3)}\right)^3 \, dx = \int x^2 \, \sec(x^3) \, dx$$

$$= \frac{1}{3} \int \sec(x^3) \, (3x^2) \, dx = \frac{1}{3} \, \ln\left|\sec(x^3) + \tan(x^3)\right| + c \, .$$

2. $\displaystyle\int \frac{2}{\sqrt{e^{4x} - 4}} \, dx$ . [2]

**Solution :** .

$$\int \frac{2}{\sqrt{e^{4x} - 4}} \, dx = \int \frac{2}{\sqrt{(e^{2x})^2 - (2)^2}} \, dx = \int \frac{e^{2x} \, (2)}{e^{2x} \sqrt{(e^{2x})^2 - (2)^2}} \, dx$$

$$= \frac{1}{2} \, \sec^{-1}\left(\frac{e^{2x}}{2}\right) + c \, .$$

3. $\displaystyle\int_0^1 10^{2\log x} \, 3^{3x^3} \, dx$ . [2]

**Solution :**

$$\int_0^1 10^{2\log x} \, 3^{3x^3} \, dx = \int_0^1 10^{\log x^2} \, 3^{3x^3} \, dx = \int_0^1 x^2 \, 3^{3x^3} \, dx$$

$$= \frac{1}{9} \int_0^1 3^{3x^3} (9x^2) \, dx = \frac{1}{9} \left[\frac{3^{3x^3}}{\ln 3}\right]_0^1 = \frac{1}{9} \left[\frac{3^{3(1)^3}}{\ln 3} - \frac{3^{3(0)^3}}{\ln 3}\right]$$

$$= \frac{1}{9} \left[\frac{3^3}{\ln 3} - \frac{3^0}{\ln 3}\right] = \frac{1}{9} \, \left(\frac{27 - 1}{\ln 3}\right) = \frac{26}{9 \ln 3} \, .$$

4. $\displaystyle\int \frac{x^2}{4 + x^2} \, dx$ . [2]

**Solution :**

$$\int \frac{x^2}{4 + x^2} \, dx = \int \frac{(x^2 + 4) - 4}{x^2 + 4} \, dx = \int \left[\frac{x^2 + 4}{x^2 + 4} - \frac{4}{x^2 + 4}\right] \, dx$$

$$= \int \left[1 - \frac{4}{x^2 + 4}\right] \, dx = \int 1 \, dx - 4 \int \frac{1}{(x)^2 + (2)^2} \, dx$$

$$= x - 4 \left(\frac{1}{2} \, \tan^{-1}\left(\frac{x}{2}\right)\right) + c = x - 2 \tan^{-1}\left(\frac{x}{2}\right) + c$$

5. $\displaystyle\int \frac{\cos\left(\ln(x^2)\right)}{x}\,dx$ . [2]

**Solution :**

$$\int \frac{\cos\left(\ln(x^2)\right)}{x}\,dx = \int \cos\left(2\ln|x|\right)\ \frac{1}{x}\,dx$$

$$= \frac{1}{2}\int \cos\left(2\ln|x|\right)\ \frac{2}{x}\,dx = \frac{1}{2}\sin\left(2\ln|x|\right) + c$$

6. $\displaystyle\int \frac{\cot\left(\sqrt[3]{x}\right)}{x^{\frac{2}{3}}}\,dx$ . [2]

**Solution :**

$$\int \frac{\cot\left(\sqrt[3]{x}\right)}{x^{\frac{2}{3}}}\,dx = \int \cot\left(x^{\frac{1}{3}}\right)\ x^{-\frac{2}{3}}\,dx$$

$$= 3\int \cot\left(x^{\frac{1}{3}}\right)\left(\frac{1}{3}\ x^{-\frac{2}{3}}\right)\,dx = 3\ln\left|\sin\left(\sqrt[3]{x}\right)\right| + c$$

7. $\displaystyle\int \frac{\left(1+\sec^{-1}x\right)^5}{\sqrt{x^4-x^2}}\,dx$ . [2]

**Solution :**

$$\int \frac{\left(1+\sec^{-1}x\right)^5}{\sqrt{x^4-x^2}}\,dx = \int \frac{\left(1+\sec^{-1}x\right)^5}{\sqrt{x^2\left(x^2-1\right)}}\,dx$$

$$= \int \frac{\left(1+\sec^{-1}x\right)^5}{|x|\sqrt{x^2-1}}\,dx = \int \left(1+\sec^{-1}x\right)^5\left(\frac{1}{x\sqrt{x^2-1}}\right)\,dx$$

$$= \frac{\left(1+\sec^{-1}x\right)^6}{6} + c \text{ , where } x > 0 \text{ .}$$

8. $\displaystyle\int \left(\sec x\ \tan x\right)\ \left(\sec^2 x\right)^{-1}\,dx$ . [2]

**First solution :**

$$\int \left(\sec x\ \tan x\right)\ \left(\sec^2 x\right)^{-1}\,dx = \int \left(\sec x\right)^{-2}\left(\sec x\ \tan x\right)\,dx$$

$$= \frac{\left(\sec x\right)^{-1}}{-1} + c = -\cos x + c$$

**Second solution :**

$$\int \left(\sec x\ \tan x\right)\ \left(\sec^2 x\right)^{-1}\,dx = \int \frac{\sec x\ \tan x}{\sec^2 x}\,dx = \int \frac{\tan x}{\sec x}\,dx$$

$$= \int \sin x\ dx = -\cos x + c\ .$$

**MATH 111 - Integral Calculus**
**Second Semester - 1446 H**
**Solution of the Second Exam**
*Dr Tariq A. Alfadhel*

**Question (1):** [4 marks]
Find $\frac{dy}{dx}$ of the following :

1. $y = \coth\left(\sqrt[3]{x}\right) + csch^{-1}\left(\frac{2}{x}\right)$ . [2]

   **Solution :**

   $$\frac{dy}{dx} = -csch^2\left(\sqrt[3]{x}\right)\ \left(\frac{1}{3}x^{-\frac{2}{3}}\right) + \frac{-1}{\frac{2}{x}\sqrt{\left(\frac{2}{x}\right)^2+1}}\ \left(\frac{-2}{x^2}\right)$$

   $$= \frac{-csch^2\left(\sqrt[3]{x}\right)}{3x^{\frac{2}{3}}} + \frac{1}{x\sqrt{\frac{4}{x^2}+1}}\ .$$

2. $y = \coth^{-1}\left(\frac{x^3}{3}\right) + \cosh^{-1}(3x)$ . [2]

   **Solution :**

   $$\frac{dy}{dx} = \frac{1}{1-\left(\frac{x^3}{3}\right)^2}\ (x^2) + \frac{1}{\sqrt{(3x)^2-1}}\ (3)$$

   $$= \frac{x^2}{1-\frac{x^6}{9}} + \frac{3}{\sqrt{9x^2-1}}\ .$$

**Question (2):** [21 marks]
Evaluate the following integrals :

1. $\int x^{-2} csch^2\left(\frac{1}{x}\right)\ dx$ . [2]

   **Solution :**

   $$\int x^{-2} csch^2\left(\frac{1}{x}\right)\ dx = \int csch^2\left(\frac{1}{x}\right)\ \left(\frac{1}{x^2}\right)\ dx$$

   $$= \int -csch^2\left(\frac{1}{x}\right)\ \left(\frac{-1}{x^2}\right)\ dx = \coth\left(\frac{1}{x}\right) + c\ .$$

2. $\int \frac{1}{\sqrt{x^2+2x+10}}\ dx$ . [2]

   **Solution :** By completing the square.

$x^2 + 2x + 10 = (x^2 + 2x + 1) + 10 - 1 = (x+1)^2 + (3)^2$ .

$$\int \frac{1}{\sqrt{x^2+2x+10}}\, dx = \int \frac{1}{\sqrt{(x+1)^2+(3)^2}}\, dx$$

$$= \sinh^{-1}\left(\frac{x+1}{3}\right) + c\ .$$

3. $\displaystyle\int (2x-1)\ \cosh x\ dx$ . [2]

**Solution :** Using integration by parts.

$u = 2x-1 \qquad dv = \cosh x\ dx$
$du = 2\ dx \qquad v = \sinh x$

$$\int (2x-1)\ \cosh x\ dx = (2x-1)\ \sinh x - \int 2\sinh x\ dx$$

$$= (2x-1)\ \sinh x - 2\int \sinh x\ dx = (2x-1)\ \sinh x - 2\cosh x + c\ .$$

4. $\displaystyle\int \cot^4 x\ \csc^4 x\ dx$ . [2]

**Solution :**

Using the substitution $u = \cot x$ .

Hence $du = -\csc^2 x\ dx \implies (-1)\ du = \csc^2 x\ dx$ .

$$\int \cot^4 x\ \csc^4 x\ dx = \int \cot^4 x\ \csc^2 x\ \csc^2 x\ dx$$

$$= \int \cot^4 x\ \left(1+\cot^2 x\right)\ \csc^2 x\ dx = \int u^4\ (1+u^2)\ (-1)\ du$$

$$= -\int \left(u^4+u^6\right)\ du = -\left[\frac{u^5}{5} + \frac{u^7}{7}\right] + c = -\frac{\cot^5 x}{5} - \frac{\cot^7 x}{7} + c$$

5. $\displaystyle\int e^{\frac{x}{2}}\ \sinh(3x)\ dx$ . [2]

**Solution :**

$$\int e^{\frac{x}{2}}\ \sinh(3x)\ dx = \int e^{\frac{x}{2}}\left(\frac{e^{3x}-e^{-3x}}{2}\right)\ dx = \int \left(\frac{e^{\frac{7x}{2}} - e^{-\frac{5x}{2}}}{2}\right)\ dx$$

$$\int \left(\frac{e^{\frac{7x}{2}}}{2} - \frac{e^{-\frac{5x}{2}}}{2}\right)\ dx = \frac{1}{2}\ \frac{2}{7}\int e^{\frac{7x}{2}}\left(\frac{7}{2}\right)\ dx - \frac{1}{2}\ \frac{2}{-5}\int e^{-\frac{5x}{2}}\left(-\frac{5}{2}\right)\ dx$$

$$= \frac{1}{7}\ e^{\frac{7x}{2}} + \frac{1}{5}\ e^{-\frac{5x}{2}} + c = \frac{e^{\frac{7x}{2}}}{7} + \frac{e^{-\frac{5x}{2}}}{5} + c\ .$$

6. $\int \frac{1}{(9-x^2)^{\frac{3}{2}}}\, dx$ . [3]

**Solution :** Using trigonometric substitutions.

Put $x = 3\sin\theta \implies \sin\theta = \frac{x}{3}$

$dx = 3\cos\theta\ d\theta$

$\left(9-x^2\right)^{\frac{3}{2}} = \left(9-9\sin^2\theta\right)^{\frac{3}{2}} = \left[9\left(1-\sin^2\theta\right)\right]^{\frac{3}{2}}$

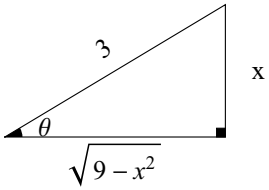
$= \left(9\cos^2\theta\right)^{\frac{3}{2}} = (9)^{\frac{3}{2}}\ \left(\cos^2\theta\right)^{\frac{3}{2}} = \left(3^2\right)^{\frac{3}{2}}\ \left(\cos^2\theta\right)^{\frac{3}{2}} = 3^3\ \cos^3\theta$

$\int \frac{1}{(9-x^2)^{\frac{3}{2}}}\, dx = \int \frac{3\cos\theta}{3^3\ \cos^3\theta}\, d\theta = \frac{1}{3^2}\ \int \frac{1}{\cos^2\theta}\, d\theta$

$= \frac{1}{9}\int \sec^2\theta\ d\theta = \frac{1}{9}\ \tan\theta + c$

$\sin\theta = \frac{x}{3}$ .

From the triangle :
$\tan\theta = \frac{x}{\sqrt{9-x^2}}$




$\int \frac{1}{(9-x^2)^{\frac{3}{2}}}\, dx = \frac{1}{9}\ \frac{x}{\sqrt{9-x^2}} + c$

7. $\int \frac{1}{\sqrt{e^{6x}+1}}\, dx$ . [2]

**Solution :**

$\int \frac{1}{\sqrt{e^{6x}+1}}\, dx = \int \frac{1}{\sqrt{\left(e^{3x}\right)^2+(1)^2}}\, dx$

$= \int \frac{e^{3x}}{e^{3x}\sqrt{\left(e^{3x}\right)^2+(1)^2}}\, dx = \frac{1}{3}\int \frac{e^{3x}\ (3)}{e^{3x}\sqrt{\left(e^{3x}\right)^2+(1)^2}}\, dx$

$= \frac{1}{3}\left(-csch^{-1}\left(e^{3x}\right)\right) + c = -\frac{1}{3}\ csch^{-1}\left(e^{3x}\right) + c$

8. $\int \tan^{-1} x\ dx$ . [3]

**Solution :** Using integration by parts.

$u = \tan^{-1} x \qquad dv = 1\ dx$

$du = \frac{1}{x^2+1}\ dx \qquad v = x$

$$\int \tan^{-1} x \; dx = x \tan^{-1} x - \int x \; \frac{1}{x^2+1} \; dx$$

$$= x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{x^2+1} \; dx = x \tan^{-1} x - \frac{1}{2} \ln(x^2+1) + c \; .$$

9. $\displaystyle \int \frac{x^2+x+9}{x^3+9x} \; dx$ . [3]

**Solution :** Using the method of partial fractions.

$$\frac{x^2+x+9}{x^3+9x} = \frac{x^2+x+9}{x(x^2+9)} = \frac{A}{x} + \frac{Bx+C}{x^2+9}$$

$$\frac{x^2+x+9}{x(x^2+9)} = \frac{A(x^2+9)}{x(x^2+9)} + \frac{x\,(Bx+C)}{x(x^2+9)}$$

$$x^2+x+9 = A(x^2+9) + x(Bx+C)$$

$$x^2+x+9 = Ax^2+9A+Bx^2+Cx = (A+B)x^2+Cx+9A$$

By comparing the coefficients of the two polynomials in each side :

$$\begin{array}{lll} A+B=1 & \longrightarrow & (1) \\ C=1 & \longrightarrow & (2) \\ 9A=9 & \longrightarrow & (3) \end{array}$$

From equation (3) : $9A = 9 \implies A = \dfrac{9}{9} = 1$ .

From equation (1) : $1 + B = 1 \implies B = 0$ .

$$\int \frac{x^2+x+9}{x^3+9x} \; dx = \int \left( \frac{1}{x} + \frac{1}{x^2+9} \right) \; dx$$

$$= \int \frac{1}{x} \; dx + \int \frac{1}{x^2+(3)^2} \; dx$$

$$= \ln|x| + \frac{1}{3} \tan^{-1} \left( \frac{x}{3} \right) + c \; .$$

**MATH 111 - Integral Calculus**
**Second Semester - 1446 H**
**Solution of the Final Exam**
*Dr Tariq A. Alfadhel*

**Question (1):** [7 marks]

1. Find the value of $c$ that satisfies the mean value theorem of the definite integral for the function $f(x) = \sqrt[3]{x-1}$ on the interval $[1, 9]$. [3]

   **Solution :** Using the formula $(b-a)\ f(c) = \int_a^b f(x)\ dx$ .

   $$(9-1)\ \sqrt[3]{c-1} = \int_1^9 \sqrt[3]{x-1}\ dx = \int_1^9 (x-1)^{\frac{1}{3}}\ dx$$

   $$8\sqrt[3]{c-1} = \left[\frac{(x-1)^{\frac{4}{3}}}{\frac{4}{3}}\right]_1^9 = \left[\frac{3}{4}\ (x-1)^{\frac{4}{3}}\right]_1^9$$

   $$8\sqrt[3]{c-1} = \frac{3}{4}(9-1)^{\frac{4}{3}} - \frac{3}{4}(1-1)^{\frac{4}{3}} = \frac{3}{4}(8)^{\frac{4}{3}} - 0 = \frac{3}{4}\left(2^4\right) = \frac{3\times 16}{4} = 12$$

   $$\implies \sqrt[3]{c-1} = \frac{12}{8} = \frac{3}{2} \implies c-1 = \frac{27}{8} \implies c = \frac{27}{8} + 1 = \frac{35}{8} \in (1, 9).$$

   The desired value is $c = \dfrac{35}{8}$ .

2. Find $F'(x)$, if $F(x) = \int_{\cos x}^{2^{3x}} \sqrt{1+t^3}\ dt$ . [2]

   **Solution :**

   $$F'(x) = \frac{d}{dx}\int_{\cos x}^{2^{3x}} \sqrt{1+t^3}\ dt$$

   $$= \sqrt{1+(2^{3x})^3}\ \left(2^{3x}(3)\ \ln 2\right) - \sqrt{1+(\cos x)^3}\ \ (-\sin x)$$

   $$= 3\ln 2\left(2^{3x}\right)\sqrt{1+2^{9x}} + \sin x\ \sqrt[3]{1+\cos^3 x}\ .$$

3. Find $y'$ if $y = \tanh\left(\sqrt{x}\right) + \sinh^{-1}\left(\dfrac{2}{\sqrt{x}}\right)$ . [2]

   **Solution :**

   $$y' = sech^2\left(\sqrt{x}\right)\ \left(\frac{1}{2\sqrt{x}}\right) + \frac{1}{\sqrt{1+\left(\frac{2}{\sqrt{x}}\right)^2}}\ \left(2\ \frac{-1}{2}\ x^{-\frac{3}{2}}\right)$$

   $$= \frac{sech^2\left(\sqrt{x}\right)}{2\sqrt{x}} - \frac{1}{x^{\frac{3}{2}}\sqrt{1+\frac{4}{x}}}\ .$$

**Question (2):** [14 marks]
Evaluate the following integrals :

1. $\int \dfrac{1}{\sqrt{-x^2-2x+8}}\,dx$ . [3]

   **Solution :** By completing the square.

   $-x^2-2x+8 = -(x^2+2x)+8 = -(x^2+2x+1)+8+1$

   $= -(x+1)^2+9 = (3)^2-(x+1)^2$ .

   $$\int \frac{1}{\sqrt{-x^2-2x+8}}\,dx = \int \frac{1}{\sqrt{(3)^2-(x+1)^2}}\,dx = \sin^{-1}\left(\frac{x+1}{3}\right)+c\ .$$

2. $\int x\ \tanh^{-1}x\ dx$ . [3]

   **Solution :** Using integration by parts .

   $u = \tanh^{-1}x \qquad dv = x\ dx$

   $du = \dfrac{1}{1-x^2}\,dx \qquad v = \dfrac{x^2}{2}$

   $$\int x\ \tanh^{-1}x\ dx = \frac{x^2}{2}\ \tanh^{-1}x - \int \frac{x^2}{2}\ \frac{1}{1-x^2}\,dx$$

   $$= \frac{x^2}{2}\ \tanh^{-1}x + \frac{1}{2}\int \frac{-x^2}{1-x^2}\,dx = \frac{x^2}{2}\ \tanh^{-1}x + \frac{1}{2}\int \frac{(1-x^2)-1}{1-x^2}\,dx\ .$$

   $$= \frac{x^2}{2}\ \tanh^{-1}x + \frac{1}{2}\int \left[\frac{1-x^2}{1-x^2} - \frac{1}{1-x^2}\right]dx$$

   $$= \frac{x^2}{2}\ \tanh^{-1}x + \frac{1}{2}\int \left[1 - \frac{1}{1-x^2}\right]dx = \frac{x^2}{2}\ \tanh^{-1}x + \frac{1}{2}\left(x-\tanh^{-1}x\right) + c$$

   $$= \frac{x^2}{2}\ \tanh^{-1}x + \frac{x}{2} - \frac{1}{2}\ \tanh^{-1}x + c$$

3. $\int \dfrac{3x^2+x+8}{x^3+4x}\,dx$ . [3]

   **Solution :** Using the method of partial fractions.

   $$\frac{3x^2+x+8}{x^3+4x} = \frac{3x^2+x+8}{x(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

   $$\frac{3x^2+x+8}{x(x^2+4)} = \frac{A\ (x^2+4)}{x\ (x^2+4)} + \frac{(Bx+C)\ x}{(x^2+4)\ x}$$

   $3x^2+x+8 = A\ (x^2+4) + (Bx+C)\ x = Ax^2+4A+Bx^2+Cx$

   $3x^2+x+8 = (A+B)x^2+Cx+4A$

By comparing the coefficients of the two polynomials in each side :

$$\begin{array}{lll} A+B=3 & \longrightarrow & (1) \\ C=1 & \longrightarrow & (2) \\ 4A=8 & \longrightarrow & (3) \end{array}$$

From equation (3) : $4A=8 \implies A=\dfrac{8}{4}=2$ .

From Equation (1) : $2+B=3 \implies B=1.$

$$\int \frac{3x^2+x+8}{x^3+4x}\,dx = \int \left(\frac{2}{x}+\frac{x+1}{x^2+4}\right)\,dx$$

$$= \int \frac{2}{x}\,dx + \int \frac{x}{x^2+4}\,dx + \int \frac{1}{x^2+4}\,dx$$

$$= 2\int \frac{1}{x}\,dx + \frac{1}{2}\int \frac{2x}{x^2+4}\,dx + \int \frac{1}{(x)^2+(2)^2}\,dx$$

$$= 2\ln|x| + \frac{1}{2}\ln\left(x^2+4\right) + \frac{1}{2}\,\tan^{-1}\left(\frac{x}{2}\right)+c\ .$$

4. $\displaystyle\int \frac{1}{(9+x^2)^{\frac{3}{2}}}\,dx$ . [3]

**Solution :** Using trigonometric substitutions.

Put $x=3\tan\theta \implies \tan\theta=\dfrac{x}{3}$ .

$dx=3\sec^2\theta\ d\theta$ .

$\left(9+x^2\right)^{\frac{3}{2}}=\left(9+9\tan^2\theta\right)^{\frac{3}{2}}=\left[9\left(1+\tan^2\theta\right)\right]^{\frac{3}{2}}=\left[3^2\ \sec^2\theta\right]^{\frac{3}{2}}$

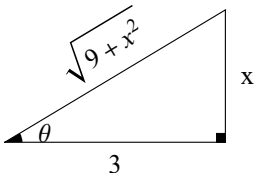
$=\left(3^2\right)^{\frac{3}{2}}\ \left(\sec^2\theta\right)^{\frac{3}{2}}=3^3\ \sec^3\theta$

$$\int \frac{1}{(9+x^2)^{\frac{3}{2}}}\,dx = \int \frac{3\sec^2\theta}{3^3\ \sec^3\theta}\,d\theta = \frac{3}{3^3}\int \frac{1}{\sec\theta}\,d\theta$$

$$= \frac{1}{9}\int \cos\theta\ d\theta = \frac{1}{9}\sin\theta + c$$

$\tan\theta=\dfrac{x}{3}$ .

From the triangle :
$\sin\theta=\dfrac{x}{\sqrt{9+x^2}}$




$$\int \frac{1}{(9+x^2)^{\frac{3}{2}}}\,dx = \frac{1}{9}\ \frac{x}{\sqrt{9+x^2}}+c\ .$$

5. $\int \frac{1}{4\ (2^{-x})+2^{x}}\ dx$ . [2]

**Solution :** .

$$\int \frac{1}{4\ (2^{-x})+2^{x}}\ dx = \int \frac{2^{x}}{2^{x}\ [4\ (2^{-x})+2^{x}]}\ dx\ .$$

$$= \int \frac{2^{x}}{4+2^{2x}}\ dx = \frac{1}{\ln 2}\int \frac{2^{x}\ \ln 2}{(2)^{2}+(2^{x})^{2}}\ dx$$

$$= \frac{1}{\ln 2}\left(\frac{1}{2}\tan^{-1}\left(\frac{2^{x}}{2}\right)\right)+c = \frac{1}{2\ \ln 2}\tan^{-1}\left(2^{x-1}\right)+c$$

**Question (3):** [19 marks]

1. Evaluate the limit $\lim\limits_{x\to\infty}\frac{x-x\ e^{x}}{x+2x\ e^{x}}$ . [2]

   **Solution :**

   $$\lim_{x\to\infty}\frac{x-x\ e^{x}}{x+2x\ e^{x}} = \lim_{x\to\infty}\frac{x\ (1-e^{x})}{x\ (1+2\ e^{x})} = \lim_{x\to\infty}\frac{1-e^{x}}{1+2\ e^{x}} \qquad \left(\frac{-\infty}{\infty}\right)$$

   Using L'Hôpital's rule.

   $$\lim_{x\to\infty}\frac{-e^{x}}{2\ e^{x}} = \lim_{x\to\infty}\frac{-1}{2} = -\frac{1}{2}$$

   Therefore, $\lim\limits_{x\to\infty}\frac{x-x\ e^{x}}{x+2x\ e^{x}} = -\frac{1}{2}$ .

2. Discus whether the improper integral $\int_{-\infty}^{\infty}\frac{1}{x^{2}+4}\ dx$ converges or diverges. [3]

   **Solution :**

   $$\int_{-\infty}^{\infty}\frac{1}{x^{2}+4}\ dx = \int_{-\infty}^{0}\frac{1}{x^{2}+4}\ dx + \int_{0}^{\infty}\frac{1}{x^{2}+4}\ dx$$

   $$= \lim_{t\to-\infty}\int_{t}^{0}\frac{1}{(x)^{2}+(2)^{2}}\ dx + \lim_{s\to\infty}\int_{0}^{s}\frac{1}{(x)^{2}+(2)^{2}}\ dx$$

   $$= \lim_{t\to-\infty}\left[\frac{1}{2}\ \tan^{-1}\left(\frac{x}{2}\right)\right]_{t}^{0} + \lim_{s\to\infty}\left[\frac{1}{2}\ \tan^{-1}\left(\frac{x}{2}\right)\right]_{0}^{s}$$

   $$= \lim_{t\to-\infty}\left[\frac{1}{2}\ \tan^{-1}\left(\frac{0}{2}\right)-\frac{1}{2}\ \tan^{-1}\left(\frac{t}{2}\right)\right]$$

   $$+ \lim_{s\to\infty}\left[\frac{1}{2}\ \tan^{-1}\left(\frac{s}{2}\right)-\frac{1}{2}\ \tan^{-1}\left(\frac{0}{2}\right)\right]$$

   $$= \left[\frac{1}{2}\ (0)-\frac{1}{2}\left(-\frac{\pi}{2}\right)\right]+\left[\frac{1}{2}\left(\frac{\pi}{2}\right)-\frac{1}{2}\ (0)\right] = \frac{\pi}{4}+\frac{\pi}{4} = \frac{\pi}{2}.$$

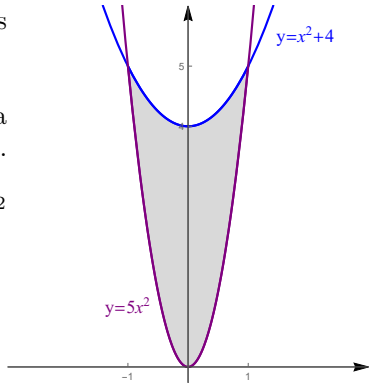
Hence, the improper integral converges.

3. Sketch the region region bounded by the curves of $y = 5x^2$ and $y = 4 + x^2$, and find its area. [3]

**Solution :**

$y = 5x^2$ represents a parabola opens upwards, and its vertex is $(0, 0)$ .

$y = x^2 + 4$ represents a parabola opens upwards, and its vertex is $(0, 4)$ .




Points of intersection of $y = 5x^2$ and $y = x^2 + 4$:
$5x^2 = x^2 + 4 \implies 4x^2 - 4 = 0$
$\implies x^2 - 1 = 0$
$\implies (x - 1)(x + 1) = 0$
$\implies x = -1 \ , \ x = 1.$

$$\mathbf{A} = \int_{-1}^{1} \left[(x^2 + 4) - (5x^2)\right] \, dx = \int_{-1}^{1} (-4x^2 + 4) \, dx$$

$$= \left[-4 \, \frac{x^3}{3} + 4x\right]_{-1}^{1} = \left[-4 \, \frac{(1)^3}{3} + 4 \, (1)\right) - \left(-4 \, \frac{(-1)^3}{3} + 4(-1)\right]$$

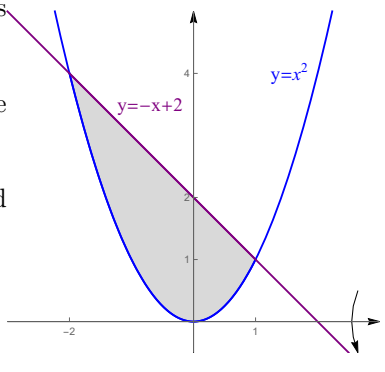
$$= -\frac{4}{3} + 4 - \left(\frac{4}{3} - 4\right) = -\frac{4}{3} + 4 - \frac{4}{3} + 4 = 8 - \frac{8}{3} = \frac{16}{3} \ .$$

4. Sketch the region bounded bu the curves $y = x^2$ and $y = -x + 2$, and find the volume of the solid generated by revolving this region about the $x-$axis. [3]

**Solution :**

$y = x^2$ represents a parabola opens upwards, and its vertex is $(0, 0)$ .

$y = -x + 2$ represents a straight line passing through $(0, 2)$ with slope $-1$.




Points of intersection of $y = x^2$ and $y = -x + 2$ :
$x^2 = -x + 2$
$\implies x^2 + x - 2 = 0$
$\implies (x + 2)(x - 1) = 0$
$\implies x = -2 \ , \ x = 1$

Using Washer Method :

$$\mathbf{V} = \pi \int_{-2}^{1} \left[ (-x+2)^2 - \left(x^2\right)^2 \right] \, dx$$

$$= \pi \int_{-2}^{1} \left[ x^2 - 4x + 4 - x^4 \right] \, dx = \pi \left[ -\frac{x^5}{5} + \frac{x^3}{3} - 2x^2 + 4x \right]_{-2}^{2}$$

$$= \pi \left[ \left( -\frac{(1)^5}{5} + \frac{(1)^3}{3} - 2(1)^2 + 4(1) \right) - \left( -\frac{(-2)^5}{5} + \frac{(-2)^3}{3} - 2(-2)^2 + 4(-2) \right) \right]$$

$$= \pi \left[ \left( -\frac{1}{5} + \frac{1}{3} - 2 + 4 \right) - \left( \frac{32}{5} - \frac{8}{3} - 8 - 8 \right) \right]$$

$$= \pi \left( -\frac{33}{5} + \frac{9}{3} + 18 \right) = \pi \left( -\frac{33}{5} + 21 \right) = \frac{72\pi}{5}$$

5. Find the arc length of the function $y = \dfrac{e^x + e^x}{2}$ on the interval $[0, \ln 3]$. [3]

**Solution :**

$$y = \frac{e^x + e^x}{2} = \cosh x$$

$$y' = \sinh x \ .$$

$$\mathbf{L} = \int_0^{\ln 3} \sqrt{1 + (\sinh x)^2} \, dx = \int_0^{\ln 3} \sqrt{1 + \sinh^2 x} \, dx$$

$$= \int_0^{\ln 3} \sqrt{\cosh^2 x} \, dx = \int_0^{\ln 3} |\cosh x| \, dx = \int_0^{\ln 3} \cosh x \, dx$$

$$= [\sinh]_0^{\ln 3} = \sinh(\ln 3) - \sinh(0)$$

$$= \frac{e^{\ln 3} - e^{-\ln 3}}{2} - 0 = \frac{3 - \frac{1}{3}}{2} = \frac{4}{3} \ .$$

6. Convert the polar equation $\dfrac{1}{r} = \dfrac{\sin\theta + 5\cos\theta}{2}$ into a Cartesian equation. [1]

**Solution :**

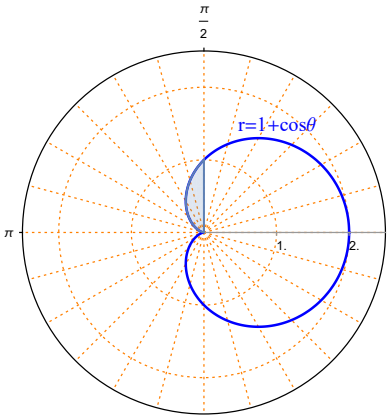
$$\frac{1}{r} = \frac{\sin\theta + 5\cos\theta}{2} \implies r(\sin\theta + 5\cos\theta) = 2$$

$$\implies r\sin\theta + 5(r\cos\theta) = 2 \implies y + 5x = 2 \implies y = -5x + 2 \ .$$

7. Sketch the region inside the graph of the polar equation $r = 1 + \cos\theta$ that is in the second quadrant. Then compute its area. [4]

**Solution :**

$r = 1 + \cos\theta$ represents a cardioid, symmetric with respect to the polar axis.




$$\mathbf{A} = \frac{1}{2}\int_{\frac{\pi}{2}}^{\pi} (1+\cos\theta)^2 \; d\theta$$

$$= \frac{1}{2}\int_{\frac{\pi}{2}}^{\pi} \left(1 + 2\cos\theta + \cos^2\theta\right) \; d\theta$$

$$= \frac{1}{2}\int_{\frac{\pi}{2}}^{\pi} \left[1 + 2\cos\theta + \left(\frac{1+\cos 2\theta}{2}\right)\right] \; d\theta$$

$$= \frac{1}{2}\int_{\frac{\pi}{2}}^{\pi} \left[\frac{3}{2} + 2\cos\theta + \frac{1}{2}\;\cos 2\theta\right] \; d\theta$$

$$\mathbf{A} = \frac{1}{2}\left[\frac{3}{2}\;\theta + 2\sin\theta + \frac{1}{4}\sin(2\theta)\right]_{\frac{\pi}{2}}^{\pi}$$

$$= \frac{1}{2}\left[\frac{3}{2}\;(\pi) + 2\sin(\pi) + \frac{1}{4}\sin(2\pi) - \left(\frac{3}{2}\;\left(\frac{\pi}{2}\right) + 2\sin\left(\frac{\pi}{2}\right) + \frac{1}{4}\sin(\pi)\right)\right]$$

$$= \frac{1}{2}\left[\left(\frac{3\pi}{2} + 0 + 0\right) - \left(\frac{3\pi}{4} + 2 + 0\right)\right]$$

$$= \frac{1}{2}\left[\frac{3\pi}{2} - \frac{3\pi}{4} - 2\right] = \frac{1}{2}\left[\frac{3\pi}{4} - 2\right] = \frac{3\pi}{8} - 1\ .$$